



# EEI4267 – Requirement Engineering

## Comprehensive Scenario-Focused Study Notes (2019 → 2024 Papers)

### 1 Exam Context & Strategy

- **Change for 2025:** No MCQs; every answer will be scenario-based essays or mini-cases.
- **Target length:** 10–20 marks → ≈1–2 written pages; 30–40 marks → ≈3–4 pages.
- **Scoring pattern:** Marks are split across (i) correct theory, (ii) application to the case, (iii) clear structure (tables, headings, use-case diagrams).

### 2 Recurring High-Value Topics

| Theme                                   | Appeared in Papers        | What You Must Demonstrate in a Scenario                                                                                             |
|-----------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Requirements hierarchy & types          | All years                 | Correctly classify Business / Stakeholder / User / System → Functional vs Non-functional vs Domain / Internal; rewrite vague lines. |
| SRS quality attributes                  | 2020-21 → 2024            | Quote IEEE 830 list and explain how to attain each attribute.                                                                       |
| RTM (Requirements Traceability Matrix)  | 2020-21 → 2024            | Define, list mandatory columns, draw sample, and describe update rhythm.                                                            |
| Gathering & elicitation techniques      | Every paper               | Name ≥ 6, choose best fit for a domain, outline execution steps.                                                                    |
| KJ / Affinity method                    | 2020-21 → 2023-24         | Give step-by-step flow for grouping ideas.                                                                                          |
| Stakeholder identification & management | All years                 | Produce justified list; map to power-interest grid.                                                                                 |
| Use-case modelling & detailed use case  | Every essay section       | Draw context, list actors, provide basic/alternate flows, assumptions.                                                              |
| Agile vs Waterfall RE                   | 2020-21 → 2023-24         | Compare roles, documents, change handling, validation timing.                                                                       |
| Project Inception / Vision              | 2020-21, 2022-23, 2023-24 | List core outputs: vision, scope, high-level risks, initial backlog or SRS outline.                                                 |
| Impact analysis & change management     | 2020-21, 2021-22, 2023-24 | Describe cost/schedule/risk assessment, Change Control Board, RTM update.                                                           |

| Theme                                 | Appeared in Papers | What You Must Demonstrate in a Scenario        |
|---------------------------------------|--------------------|------------------------------------------------|
| Validation vs Verification            | 2021-22, 2022-23   | Provide definitions, techniques, deliverables. |
| Non-functional requirement categories | All papers         | List category, give measurable fit-criterion.  |

### 3 Reference Sheet for Key Concepts

#### 3.1 Requirement Types & Definitions

| Level          | Definition                                     | Example (Ticketing System)                           |
|----------------|------------------------------------------------|------------------------------------------------------|
| Business       | High-level goal or benefit to the organisation | "Increase ticket-sale revenue by 15% within a year." |
| Stakeholder    | Need expressed by any stakeholder group        | "Finance team needs daily sales exports to ERP."     |
| User           | What end users must achieve                    | "Fan must download e-ticket after payment."          |
| System         | What the software must provide                 | "System shall generate PDF ticket with QR code."     |
| Functional     | Observable service or behaviour                | Seat selection, payment capture.                     |
| Non-Functional | Quality or constraint on behaviour             | Page load < 3 s, 256-bit TLS for payments.           |
| Domain         | Industry-specific rule                         | Under-18 users cannot buy liquor in stadium.         |
| Internal       | Development-team concerns                      | "Code shall follow PEP-8."                           |

#### 3.2 Good SRS Checklist (FUCC-TV)

| Attribute                                       | Practical Technique                                |
|-------------------------------------------------|----------------------------------------------------|
| <b>F</b> unctional & non-functional items split | Use separate numbered sections                     |
| <b>U</b> nambiguous                             | Define all acronyms; use "shall" not "should"      |
| <b>C</b> omplete                                | Trace each user story to at least one requirement  |
| <b>C</b> onsistent                              | Single source of truth for terms; RTM cross-checks |
| <b>T</b> raceable                               | Unique IDs, RTM linking to design/test cases       |
| <b>V</b> erifiable                              | Add measurable acceptance criteria                 |

### 3.3 Requirements Traceability Matrix (RTM) Sample

| Req-ID | Description                                  | Type            | Priority | Source           | Design Ref    | Test-Case     | Status |
|--------|----------------------------------------------|-----------------|----------|------------------|---------------|---------------|--------|
| FR-07  | System shall prevent double-booking of seats | Functional      | High     | UC-03            | Class SeatMgr | TC-18         | Pass   |
| NFR-02 | Page load $\leq 3$ s for 95% of requests     | Performance NFR | High     | Ticketing Vision | NFR-Spec§2    | Perf-Suite-05 | Fail   |

*Update cadence:* every approved change request triggers RTM revision before design or code starts.

### 3.4 Common Gathering & Elicitation Techniques

| Technique                  | Best-fit Scenario                   | Strength           | Pitfall                        |
|----------------------------|-------------------------------------|--------------------|--------------------------------|
| Semi-structured interviews | Domain experts, prison guards       | Deep insights      | Time-intensive                 |
| Workshops/JAD              | Multi-stakeholder alignment         | Rapid consensus    | Needs skilled facilitator      |
| Surveys                    | Large user bases                    | Wide coverage      | Low depth                      |
| Observation (shadowing)    | Operational processes               | Contextual reality | Hawthorne effect               |
| <b>KJ / Affinity</b>       | Early idea reduction                | Removes duplicates | Needs follow-up prioritisation |
| Prototyping                | UI-heavy apps                       | Early validation   | Scope creep                    |
| Document analysis          | Re-engineering (PrisTrack, railway) | Baseline knowledge | Outdated docs                  |

### 3.5 Validation vs Verification

| Aspect     | Validation ("Are we building the right product?") | Verification ("Are we building the product right?")         |
|------------|---------------------------------------------------|-------------------------------------------------------------|
| Timing     | Before/throughout development                     | Throughout + post-implementation                            |
| Techniques | Reviews, walkthroughs, prototypes, UAT            | Inspections, unit/integration/system tests, static analysis |
| Output     | Stakeholder sign-off                              | Test reports, defect logs                                   |

### 3.6 Agile-Waterfall Comparison (Requirements View)

| Point    | Waterfall               | Scrum / Agile            |
|----------|-------------------------|--------------------------|
| Baseline | Fixed SRS before design | Evolving product backlog |

| Point           | Waterfall                                                          | Scrum / Agile                                      |
|-----------------|--------------------------------------------------------------------|----------------------------------------------------|
| Change handling | Formal CR + impact analysis                                        | Backlog re-prioritisation each sprint              |
| Documentation   | Heavy upfront (SRS, RTM)                                           | "Just enough" – user stories, definition of done   |
| Traceability    | Document-centric RTM                                               | Tool-assisted (Jira links stories → tasks → tests) |
| Similarities    | Need stakeholder involvement, acceptance criteria, version control |                                                    |

## 4 Core Processes & Templates for Scenario Answers

### 4.1 Requirement Gathering Road-Map (Waterfall)

1 Identify stakeholders → 2 Select techniques → 3 Plan sessions & artefacts → 4 Conduct sessions → 5 Document in SRS → 6 Review & validate.

### 4.2 User Story Format & Checklist

As a <role> I want <goal> so that <value>.

- **INVEST** qualities: Independent, Negotiable, Valuable, Estimable, Small, Testable.

### 4.3 Detailed Use-Case Template

| Field                | Guidance                              |
|----------------------|---------------------------------------|
| Name                 | Verb + Noun ("Book Seat")             |
| Primary Actor        | e.g., Customer                        |
| Goal                 | Complete ticket purchase              |
| Pre-conditions       | Seat map loaded, user logged in       |
| Basic Flow           | Steps 1-N                             |
| Alternate Flows      | AF-1: payment failed; AF-2: seat sold |
| Post-conditions      | Ticket reserved, e-ticket emailed     |
| Non-functional notes | Response ≤ 5 s                        |

### 4.4 Impact-Analysis Grid (PAIR)

| P | People affected | Roles needing training |  
| A | Architecture | Modules touched |  
| I | Interfaces | Upstream/downstream contracts |  
| R | Risks | Cost, schedule, quality impacts |

## 5 Domain Practice Sets

Prepare mini-artefacts (stakeholder list, requirement classification, RTM snippet, use-case diagram) for:

1. **Cinema / Stadium ticketing** (2019-20, 2023-24)
2. **Prisoner tracking biometrics** (2020-21)
3. **Rail-elephant collision avoidance mobile app** (2021-22)
4. **Online event management platform** (2022-23)

## 6 Tooling Cheat-Sheet

| Purpose          | Lightweight Tool                                   | Enterprise Tool    |
|------------------|----------------------------------------------------|--------------------|
| Backlog & boards | Trello, GitHub Projects                            | Jira, Azure DevOps |
| Diagramming      | <a href="https://draw.io">draw.io</a> , Lucidchart | Sparx EA           |
| RTM maintenance  | Excel w/ IDs & hyperlinks                          | Jama, DOORS NG     |
| Prototyping      | Figma                                              | Axure, Balsamiq    |

## 7 Typical Essay Prompts & Model Openings

| Prompt                                                 | Recommended First Sentence                                                                                                                                                                                                    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Explain RTM to the team."                             | "A Requirements Traceability Matrix is a living document that maps each stakeholder need to design, code, and test artefacts, ensuring no requirement is overlooked during development or change."                            |
| "Separate functional and non-functional requirements." | "Functional requirements describe observable system behaviours, whereas non-functional requirements constrain the manner in which these behaviours are delivered; applying these definitions, the list divides as follows..." |
| "Compare Scrum and Waterfall RE."                      | "Both Scrum and Waterfall perform the five RE activities—elicitation, analysis, documentation, validation, and management—but they differ in timing, artefacts, and change handling as summarised in Table 1."                |

## 8 Exam-Day Rapid-Recall Mnemonics

- **FUCC-TV** – good SRS.
- **INVEST** – good user story.
- **PAIR** – impact analysis dimensions.
- **SMART** – measurable NFR fit-criteria.

## 9 Self-Check Before the Exam

- ☐ Can I draft a full-dress use-case in < 7 minutes?
- ☐ Can I explain RTM purpose and draw a 6-column sample from memory?
- ☐ Do I know at least four elicitation techniques and when each is ideal?
- ☐ Can I write two crystal-clear user stories and confirm INVEST?
- ☐ Am I able to quote the six key SRS qualities without reference?
- ☐ Can I articulate two similarities and two differences between Agile and Waterfall RE?

## Final Tip

Write every answer in **three layers**:

1. **Definition/Principle** – proves theory mastery.
2. **Application to the given case** – shows relevance.
3. **Artefact or table** – earns structure marks and demonstrates professional practice.

Master these layers and the recurring themes above, and scenario-based questions in EEI4267 become straightforward. Good luck!

